Boxuan Wang

Design Engineering Student

Robotics | Intelligent Systems | jw4424@ic.ac.uk | mystify59.github.io

PROFILE

Design Engineering student at Imperial College London with experience in humanoid development, robot testing and Python, plus team work in robot vision and autonomous control.

EDUCATION

Imperial College London 2024 - 2028

Design Engineering | Expected completion 2028

Relevant study: Intelligent Systems, Physical Computing, Mechanical Engineering, FEA.

The King's School, Canterbury 2019 - 2024

King's Scholar; Vice Captain of Fencing.

WORK EXPERIENCE

Li Auto, *Humanoid Development Intern* July - September 2026

- Collected human-motion data using wearable trackers, VR headsets and robot arms mirroring operators for tasks including parts organisation.
- Designed connector parts and structural supports in SolidWorks, supported by Ansys analysis.
- Redesigned upper-torso load paths as part of the team, reducing torso weight by **18%** and meeting fatigue requirements in Ansys simulations with a safety factor of 3.

Elibot, *Cooperative Robot Intern* July - August 2023

- Created joint-specific fatigue tests, logged faults and examined angular differences during braking in harmonic-drive joints.
- Built a Python system to log incoming, produced and shipped robots.
- Contributed to app and web interfaces for remote robot control and webpage UX.

SELECTED PROJECTS

Autonomous Football Robot, *Team robotics project* 2025

- Contributed to team model training and ball-tracking code using OpenMV and Edge Impulse.
- Contributed to team control-code development for autonomous navigation and task execution on an existing robot platform.

Street Pulse, *Electronics prototyping and testing* 2025

- Prototyped electronics, assembled and tested a vibration module for a team interactive-floor project.

SUV Aerodynamics and Battery Cooling, *Thermofluids project* 2026

- Built SUV geometry in Fusion and a 400,000-cell Ansys Fluent model to analyse pressure distribution and rear-wake behaviour.
- Contributed to a team cold-plate battery-cooling design, evaluating flow distribution, heat transfer and thermal-performance trade-offs.

SKILLS

Programming and vision. Python, C++, MATLAB; OpenMV, Edge Impulse.

Mechanical design. SolidWorks, Fusion; CAD, design for additive manufacturing, prototype assembly.

Analysis and testing. Ansys Mechanical, Fluent; FEA, CFD, fatigue analysis and robot-joint testing.

AI tools. AI agents and MCP tools for 3D-printable validation structures and temporary prototypes.

LANGUAGES

Chinese (Native) | **English** (Fluent) | **French** (Basic)